

[ADR006] Oracle CDC Solution Selection for TMS Branch Databases

Status: Accepted

Date: 2026-04-15

Decision Date: 2026-04-28

Context

The New Dispo system requires Change Data Capture (CDC) from on-premises Oracle TMS databases to enable real-time data synchronization for TMS branch operations. The existing CDC pipeline (ADR-001) uses Google Datastream for Postgres-based branches. With the strategic shift toward Oracle TMS at branches, a CDC solution is needed that supports Oracle as a source and delivers change events to GCP Cloud Storage.

Stakeholders: P3 (Matthias Max, Martin Dittmann), Nagel IT (Christian Lang, Matt Wilkinson, Ron Vervenne, Thomas Paulus)

Go-Live Target: June 2026

A parallel PoC was executed from March to April 2026, testing both Striim and Google Datastream against Oracle TMS1060.SENDUNG (UAT environment). The PoC scope ended at Cloud Storage — downstream processing was considered post-PoC.

Options Considered

- **Option A: Striim** — extend existing Nagel Striim deployment; self-managed VM-based platform hosted in GCP WL3. Sub-second latency proven.
- **Option B: Google Datastream** — GCP-native serverless CDC service; consistent with existing Postgres CDC pattern. Latency depends on Oracle redo log configuration and Datastream read mode:
 - **B1: Datastream + Oracle redo log tuning** (ARCHIVE_LAG_TARGET=900, 256MB logs) — expected 5-20 min latency, low effort
 - **B2: Datastream + Binary Log Reader** (Preview, not GA) — expected 1-5 min latency, no SLA
 - **B3: Datastream as-is** (current Oracle config) — 42-66 min latency, unacceptable

Decision

Option A: Striim — accepted for Go-Live 1060 (June 2026).

Decided in the April 28, 2026 follow-up meeting. Striim is the CDC solution for the go-live timeline. The Datastream Binary Log Reader (Option B2) is being explored in parallel as a potential post-go-live optimization but is not blocking the go-live decision. The Datastream LogMiner approach (Option B3) was ruled out on April 21 due to unacceptable latency (~42-66 min).

Rationale for Striim over waiting for Binary Log Reader:

- Striim is production-proven (~2 years streaming Oracle to GCP at Nagel)
- Sub-second latency confirmed in PoC (vs. Binary Log Reader's unproven performance at scale)
- Striim license secured until October 2026 (borrowed Google license, extended by Matt Wilkinson)
- Binary Log Reader is in GCP Preview (not GA), requires excessive Oracle permissions beyond documentation, and has not been validated under production load
- Go-live deadline (June 2026) does not allow waiting for Binary Log Reader GA or further investigation

Rationale

PoC Results Summary

Criterion	Striim	Datastream
End-to-end latency	~100ms (flat, no variance)	~42-66 min avg (P50). Initial reports of 16-20s were system processing latency only — 99.4% of total latency is read lag waiting for Oracle archived redo logs
Completeness	99.98% (60,967 of 60,978 events)	100% (23,751 records, zero errors)
Managed service	No (self-managed VM)	Yes (serverless)
GCP-native	No	Yes

Patrick Uschmann (PO) verbally confirmed on 2026-04-08 that **~10 seconds is acceptable; minutes would be problematic**. This tolerance is not yet formally documented and needs re-evaluation given the corrected latency numbers.

Costs

	Datastream	Striim (borrowed license)*	Striim (commercial 8-core min.)*
Monthly (64 DBs)	EUR 344	EUR 2,771*	EUR 20,571*
Annual (64 DBs)	EUR 4,124	EUR 33,248*	EUR 246,848*
Ratio	1x	8x	60x

** Striim is currently running on a borrowed Google license (confirmed unsustainable by Matt Wilkinson). If Nagel must pay independently, marketplace pricing starts at \$19,200/month (8-core minimum tier). Actual Striim quote required.*

Consequences

If Striim is Selected

- **Positive:** Sub-second latency; 99.98% delivery rate; already deployed and operational; fastest path to go-live
- **Negative:** Unknown and potentially significant licensing costs (\$19,200+/month); not GCP-native; self-managed VM infrastructure

If Datastream is Selected (with Oracle tuning)

- **Positive:** GCP-native; serverless; transparent pricing (~EUR 344/month for 64 DBs); consistent with existing Postgres CDC pattern
- **Negative:** Higher latency (5-20 min even after tuning); requires Oracle DBA coordination; second PoC needed to validate

Related ADRs

- [\[ADR001\] Data Exchange Between TMS and CALSuite's Cross-Dock](#) — established Datastream as CDC solution for Postgres
- [\[ADR007\] Retain PSC Proxy for Datastream-to-AlloyDB Connectivity](#) — proxy decision for Postgres Datastream

References

- Detailed analysis: [Datastream CDC Analysis Report](#)
- Project status: [Oracle CDC Project](#)
- Azure DevOps: [Feature 121925: TMS Pulse ORA Extension](#)

Document History

Date	Author	Change
2026-04-15	Matthias Max	Initial ADR created from PoC results
2026-04-17	Matthias Max	Complete rewrite: incorporated deep analysis (CW 16), corrected latency from 16-20s to 42-66 min, slimmed format
2026-05-11	Matthias Max	ADR accepted: Striim selected (decided Apr 28 meeting). Decision rationale added. Binary Log Reader noted as parallel post-go-live track